GEO-Insight

Overlookedness as Disagreement: A Multi-Attribute, Multi-Stakeholder Gap-Ranking System for Humanitarian Crises

Cache Me if You Can — Datathon 2026

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Abstract

Humanitarian coordinators and donor advisors need rapid, defensible answers to one question: *where does documented need most outpace available funding?* The 2026 Global Humanitarian Overview asks $33B for 135M people, yet the 2025 appeal raised only $12B — the lowest in a decade. Nineteen country contexts are now severely underfunded, up from eight in 2021. When almost every response is underfunded, the useful question shifts from *which crises are funded?* to *relative to documented need, which crises are most overlooked?*

Existing tools — CERF UFE rankings, FTS coverage ratios, and Palantir-Foundry-based systems such as WFP's DOTS — treat overlookedness as a **scalar**: one number per country-year. Our contribution is a *joint-distribution* view of overlookedness, anchored in three explicit research gaps in the 2015–2026 humanitarian-analytics literature. We extend the MADM/AHP/MAUT framework of Rye & Aktas [1] with two new axes: (i) an **intra-crisis equity term** measuring inequality of coverage across sector clusters within a single country, following the fair-distribution formulation of Vargas Florez et al. [4]; (ii) an **inter-stakeholder disagreement term** measuring rank variance across four donor-preference profiles, operationalising the per-decision-maker rank divergence empirically demonstrated by Rye & Aktas (Table 13). The resulting four-cell typology — *consensus-overlooked*, *contested*, *sector-starved*, or combinations — is a classification that current tools conflate. Validated against CERF Underfunded Emergencies (consensus axis) and CARE *Breaking the Silence* (sector-starved axis).

1 Motivation

The humanitarian system in 2026 is operating under unprecedented fiscal strain. According to OCHA's Global Humanitarian Overview 2026,[1] agencies are asking $33B to reach 135 million people, with a hyper-prioritised $23B floor targeting 87 million lives. The 2025 appeal raised only $12B. Headline coverage ratios are collapsing: DRC ∼22%, Yemen and Somalia ∼24%, Sudan from 69% (2024) to 35% (2025).[2] OCHA now flags 19 contexts as severely underfunded (below 50% coverage), up from eight in 2021 — severe underfunding is the norm, not the exception.

The analytical implication: scoring overlookedness with a single coverage ratio is no longer discriminative. When 19 contexts are all under 50% coverage, the interesting variation is no longer *whether* a crisis is underfunded but *how* and *for whom*.

2 Problem Definition

Given a query or geographic scope (country, region, sector, or natural-language prompt), GEO-Insight returns a ranked list of crises. Each ranked entry carries:

- a **median gap score** across four donor-preference profiles;

[1] https://www.unocha.org/publications/report/world/global-humanitarian-overview-2026-enesfr
[2] https://www.thenewhumanitarian.org/news/2025/12/08/five-takeaways-uns-aid-plans-2026

- an **inter-profile disagreement band** (rank inter-quartile range across profiles);
- an **intra-crisis equity score** (cluster-coverage Gini);
- a **four-cell typology label** (Section 5.8);
- an attribute-level decomposition of the score;
- data-quality flags (stale HNO, incomplete funding data, no active HRP);
- a brief natural-language explanation suitable for a briefing note.

Ambiguous queries are handled by graceful degradation. Outputs do not fabricate figures absent from the source data. Uncertainty is surfaced rather than suppressed.

3 Related Work and the Gap We Close

Three strands of literature shape the methodology. The synthesis below emphasises *convergent language across papers* about what the field is missing.

Multi-attribute decision-making for humanitarian allocation. Rye & Aktas [1] propose EDIS, a decision-support framework using multi-attribute decision making (MADM), AHP for preference elicitation, and MAUT for aggregation. Their Table 13 explicitly demonstrates supplier-rank divergence under different decision-maker preferences — but stops short of aggregating this divergence into a meta-metric. Their Section 7 flags "actual non-resource-based determinants of supplier selection" as open.

Three papers converge on equity as the open gap. Abdulrashid et al. [2], in a 2026 systematic review of AI-DSS for disaster management, state (§7.3): *"Despite the growing recognition that humanitarian logistics decisions often have ethical implications, few studies incorporate equity-aware or fairness-oriented modelling."* Table 12 lists "How can recovery-focused AI-DSS incorporate equity metrics into rebuilding prioritisation?" as one of only five items in its research agenda.

Sahebjamnia et al. [3], concluding their hybrid-DSS design, name the same gap (§6, p. 22): *"using more performance measures . . . involving several egalitarian and utilitarian measures is of vital importance."*

Vargas Florez et al. [4] give the equation template. Their Eqs. 9–12 introduce a fair-distribution penalty on the maximum regional shortage percentage $PM(s) = \max_i R(i,s)/d(i,s)$, where R is unmet requirement and d is demand. The efficiency objective, they note explicitly, *"conflicts with the criterion of fair distribution."* We generalise this formulation from inter-regional to inter-cluster inequality within a single country response.

Embedded, interpretable analytics. Griffith et al. [5] argue that during time-constrained humanitarian operations, simple and interpretable analytics outperform complex black-box methods, and characterise the field as "data rich, information poor" with data integration (not algorithm choice) as the operational bottleneck. We take this as a constraint on any method that adds to the score vector: every new term must decompose cleanly and be explainable to a non-technical analyst.

What GEO-Insight adds that is not in this literature. No paper in our review aggregates per-decision-maker rank divergence into an operational "disagreement" meta-score. No paper operationalises intra-crisis equity as a Gini-type inequality on cluster coverage ratios within a single country HRP. Both are small, defensible extensions of existing frameworks. Neither is addressed by Palantir-Foundry-based systems such as WFP's DOTS [6], which unify an

organisation's operational data toward a single source of truth — the opposite of surfacing inter-stakeholder disagreement.

4 Data Sources

Dataset	Role	Source
HNO (Humanitarian Needs Overview)	People in need by country *and sector*	HDX / OCHA HPC Tools
HRP (Humanitarian Response Plans)	Plan targets, status, requirements	HDX / OCHA
FTS Requirements & Funding	Requested, pledged, received funding (with sector breakdown)	HDX / OCHA FTS
CoD-PS (Global population)	Denominator for per-capita normalisation	HDX
CBPF Pooled Funds Data Hub	Country-based pooled fund allocations	cbpf.data.unocha.org
HDX HAPI	Unified daily-refresh API over above sources	HDX
CERF Underfunded Emergencies	**Validation benchmark** (consensus axis)	cerf.un.org
CARE *Breaking the Silence*	**Validation benchmark** (sector/attention axis)	ReliefWeb annual report
INFORM Severity Index (required for a_4)	Monthly 1–10 severity, 109 countries, 2020-09–2026-03	EU JRC DRMKC
IPC (backup for a_4 on food-security crises)	Food-security phase classification 1–5	ipcinfo.org
ACLED, UNHCR (optional enrichment)	Conflict events, displacement	External, declared

Table 1: Primary data sources. Sector-level FTS breakdown is load-bearing for the intra-crisis equity term; country-level only collapses the methodology.

4.1 Attribute-to-source mapping

An audit of the five downloaded HDX datasets (run 2026-04-18) maps each score attribute to a concrete file and column set. Five of six attributes are computable directly from downloaded data; one (severity) requires a single external fetch.

4.2 Gaps and public-source workarounds

Three items needed by the methodology are not on disk. All three are openly available online.

- **Severity (a_4) — solved at global scale via INFORM Severity.** HNO 2024/2025 CSVs do not carry a structured severity score. We therefore fold in the EU JRC's INFORM Severity Index from https://drmkc.jrc.ec.europa.eu/inform-index/INFORM-Severity/Results-and-data: a 1–10 composite assessed monthly for every active humanitarian crisis worldwide. All 67 monthly snapshots (2020-09 to 2026-03) are downloaded by `Data/Third-Party/DRMKC-INFO` and consolidated by `consolidate.py` into a single long-format CSV with 5,298 (country × month) rows covering 109 countries; every full-stack priority country has ≥60 months of history. This is load-bearing in two ways: it resolves the a_4 attribute globally, and it supplies the panel-data backbone that makes the temporal bonus task tractable (state-space filtering, hierarchical baselines). An earlier IPC export remains at `Data/Third-Party/IPCInfo/` as a richer single-country backup (Afghanistan, 17 analyses 2017–2025) for demo purposes.

- **CERF UFE benchmark.** Annual list published on CERF's website; scraped manually once per cycle. ∼30 entries per year, trivial to transcribe.
- **CARE *Breaking the Silence*.** Annual PDF on CARE and ReliefWeb; top-10 list is extracted manually.

4.3 Required pre-processing

Two concrete engineering steps are required before the scoring pipeline can run.

- **Cluster-taxonomy harmonisation.** HNO and FTS label identical sector clusters inconsistently — HNO "Sanitation & Hygiene" vs. FTS "Water Sanitation and Hygiene" vs. the French "EHA - Eau Hygiène Assainissement." We maintain a manual mapping table of ∼20–30 entries at `src/taxonomies/cluster_map.csv` to canonicalise cluster names before joining on `(country, cluster, year)`. Without this mapping, a_6 collapses silently.
- **Multi-country FTS allocation for HHI.** Some FTS incoming records have pipe-delimited `destLocations` spanning multiple countries (regional plans). We apply the following rule: if the destination plan code unambiguously identifies a lead country (e.g. `HAFG25` → AFG) the funding is attributed to that country; otherwise the amount is split equally across listed destinations and the resulting a_5 is flagged "estimated."

4.4 Verdict

All six score attributes and both validation benchmarks are recoverable from public sources. Severity is the only attribute requiring an external fetch; IPC provides it. No attribute is non-computable. **The methodology is feasible under the available data** — the binding constraint is engineering time on the two pre-processing tasks above plus one IPC fetch, not data existence.

5 Methodology

5.1 Unit of analysis

The primary unit is the *country-level crisis response* for an active HRP cycle. The intra-crisis equity term (Section 5.4) requires sector-level decomposition; where HNO/FTS sector breakdowns are absent or stale, the equity term degrades to a null value and is flagged rather than imputed.

5.2 Eligibility filtering

A crisis u enters the ranking pool only if: (1) an active HRP exists *or* documented PIN exceeds τ_{PIN} (default 10^6); (2) HNO figures are no older than τ_{stale} months (default 24); (3) at least one FTS funding record exists for the current or prior plan year. Crises failing (1) are reported in a secondary "possible blind-spot" list; not scored against HRP-backed crises.

5.3 Attribute vector (six attributes)

For each eligible crisis u, let $P(u)$ denote people in need, $R(u)$ funding requirements, $F(u)$ funding received, $N(u)$ country population, $D(u)$ the vector of per-donor contributions, and for each sector cluster c, let $r_c(u) = F_c(u)/R_c(u)$ be the cluster coverage ratio and $P_c(u)$ be cluster-level PIN. Then:

$$
\begin{aligned}
a_1(u) &= 1 - \min(1,\ F(u)/R(u)) && \text{coverage shortfall} \\
a_2(u) &= \max(0,\ R(u) - F(u))/P(u) && \text{per-PIN funding gap (USD)} \\
a_3(u) &= P(u)/N(u) && \text{need intensity} \\
a_4(u) &= \text{severity}(u) \in [1,5] && \text{HNO severity / IPC phase} \\
a_5(u) &= \text{HHI}(D(u)) && \text{donor concentration} \\
a_6(u) &= G_{\text{cluster}}(u) && \text{intra-crisis equity (new)}
\end{aligned}
\tag{1}
$$

5.4 Intra-crisis equity term a_6

Following the fair-distribution logic of Vargas Florez et al. [4] (Eqs. 9–12), we measure inequality of coverage across sector clusters within a single country's response. Let $\mathcal{C}(u)$ be the set of clusters with non-null FTS and HNO figures. The PIN-weighted Gini coefficient of coverage is:

$$
G_{\text{cluster}}(u) \;=\; \frac{\sum_{i\in\mathcal{C}}\sum_{j\in\mathcal{C}} P_i(u)\,P_j(u)\,|r_i(u)-r_j(u)|}{2\left(\sum_{c\in\mathcal{C}} P_c(u)\right)^2\,\bar{r}(u)} \qquad \bar{r}(u) = \frac{\sum_c P_c(u)\,r_c(u)}{\sum_c P_c(u)} \tag{2}
$$

$G_{\text{cluster}}(u) \in [0,1]$ with 0 meaning all clusters equally funded and 1 meaning total concentration. A crisis with aggregate coverage 80% but a Health cluster at 20% carries a large a_6 and is flagged as *sector-starved*. This directly operationalises the "egalitarian measures" gap named by Sahebjamnia et al. [3] (§6) and Abdulrashid et al. [2] (§7.3, Table 12).

An alternative Vargas-Florez-faithful formulation tracks the maximum per-PIN shortage across clusters:

$$
a_6'(u) = \max_{c\in\mathcal{C}(u)} (R_c(u) - F_c(u))/P_c(u).
$$

We treat a_6' as a sensitivity check; the Gini is the primary choice because its $[0,1]$ bound makes it directly compatible with the MAUT utility transform.

5.5 Normalisation and utility

Each attribute is min-max normalised over the eligible pool to $\tilde{a}_i(u) \in [0,1]$, oriented so that larger values indicate "more overlooked." Per-attribute utility U_i is linear by default; for coverage shortfall (a_1) we use concave $U_1(x) = 1-(1-x)^\gamma$, $\gamma \in [1.5, 2.5]$, to amplify the low-coverage tail (moving from 10% to 5% coverage matters more than moving from 60% to 55%).

5.6 Four donor-preference profiles

Overlookedness is computed under four distinct weight vectors $w_p \in \Delta^5$, one per stylised donor profile. Weights are defaults, not claims about any specific institution's formal policy; they are tuneable and serve to expose how the ranking shifts under plausible mandates.

5.7 Composite gap score and disagreement band

Under profile p the score is:

$$
G_p(u) \;=\; \sum_{i=1}^{6} w_{p,i}\,U_i(\tilde{a}_i(u)) \tag{3}
$$

For each crisis u we then rank the pool under every profile and report:

- $\text{rank}_p(u)$ for each $p \in \{\text{CERF, ECHO, USAID, NGO}\}$;
- $\widetilde{\text{rank}}(u) = \text{median}_p\,\text{rank}_p(u)$;
- $\text{IQR}(u) = \text{IQR}_p\,\text{rank}_p(u)$ — the *disagreement band*.

Small IQR on a high median rank indicates *consensus overlooked*: every plausible donor mandate ranks this crisis highly. Large IQR indicates *contested*: the crisis's priority depends on the weighting philosophy.

5.8 Four-cell typology

Combining the disagreement band with the intra-crisis equity term yields a four-cell typology (medians split the eligible pool into quartiles; "high" means top-quartile):

5.9 Explanation layer

The per-attribute contribution $c_{p,i}(u) = w_{p,i}\, U_i(\tilde{a}_i(u))$ decomposes the score under each profile (Griffith et al. [5] interpretability principle). For each top-K crisis we report the dominant two attributes per profile, absolute reference figures (PIN, requirements, received, coverage), typology label, and data-quality caveats.

5.10 Query layer

Natural-language queries parse via a lightweight LLM call with fixed prompt and JSON schema. *"Acute food insecurity hotspots with below 10% funding"* $\Rightarrow$ `{sector: "food", severity:` ≥ 4`, coverage_max: 0.10}`. The LLM is not involved in scoring. Schema-validation failures fall back to keyword filtering with a degradation flag.

6 Third Axis: Structural vs. Acute Neglect (Bonus)

The typology in Section 5.8 is a point-in-time view. To address the bonus question — distinguishing chronic from recent underfunding — we add a third, temporal axis. Let $G_{p,t}(u)$ be the profile-p score computed using data available at year t. For a rolling window of k years:

$$\bar{G}_{p,k}(u) = \frac{1}{k} \sum_{t=T-k+1}^{T} G_{p,t}(u) \qquad \text{chronic component} \tag{4}$$

$$\Delta G_p(u) = G_{p,T}(u) - \frac{1}{k-1} \sum_{t=T-k+1}^{T-1} G_{p,t}(u) \qquad \text{acute component} \tag{5}$$

We average these over profiles and surface the (chronic, acute) pair as a 2-D view per crisis. A crisis in the *consensus-overlooked, high-chronic* quadrant is a candidate for multi-year pooled funding; one in *consensus-overlooked, high-acute* is a candidate for the CERF rapid-response window. This explicitly connects the policy recommendation to the ranking output.

7 System Architecture

1. **Ingestion.** HDX HAPI client (preferred) with fallback to direct CSV pulls. Pandas dataframes cached locally with dataset version and fetch timestamp.
2. **Harmonisation.** Country ISO-3 canonicalisation, sector taxonomy mapping (HNO clusters $\leftrightarrow$ FTS sector codes), plan-year alignment.
3. **Scoring engine.** Pure-Python computation of Equations (1)–(3); deterministic, fully unit-testable. Gini via a vectorised pandas implementation.
4. **Multi-profile driver.** Re-runs scoring under each w_p; aggregates $\widetilde{\text{rank}}$ and IQR.
5. **Query layer.** Pydantic schema + LLM-assisted parser; fallback to keyword routing.

6. **Presentation.** Notebook for development; Streamlit dashboard with four-cell scatter plot; JSON export.

8 Validation Strategy

We validate against *two independent benchmarks* — one per axis.

- **Consensus axis (CERF UFE).** Our *consensus-overlooked* list should overlap substantially with OCHA's CERF Underfunded Emergencies window selections.[3] Metrics: Overlap@10, Spearman ρ between $\widetilde{\text{rank}}$ and CERF UFE inclusion order.
- **Sector-starved axis (CARE Breaking the Silence).** Our high-G_{cluster} list should correlate with the top under-reported crises catalogued in CARE International's annual report.[4] Metrics: Overlap@10, rank correlation.
- **Contested axis (qualitative).** The *contested* list is hard to validate quantitatively but should align with crises known to exhibit donor tension (e.g. Venezuela, Ethiopia–Tigray, Myanmar). We report this as a qualitative cross-check, flagged as such.
- **Sensitivity.** How much does top-K shift under leave-one-profile-out? Low shift indicates the consensus set is not an artefact of profile choice.
- **Stress tests.** Deliberate removal of sectors, years, or donor data to confirm graceful degradation and correct data-quality flags.

Two independent benchmarks on two orthogonal claims (consensus underfunding, and intra-crisis cluster inequality) materially strengthens the jury story over single-benchmark validation.

9 Handling Uncertainty and Data Quality

Following Abdulrashid et al. [2], we treat data-quality transparency as a first-class output.

- **Stale HNO figures.** Vintage carried through scoring; crises with PIN >24 months old flagged amber.
- **Funding definition.** We commit to *funding received* as the numerator of coverage; all three (requested/pledged/received) reported in the per-crisis detail view.
- **Missing sector data.** Equity term degrades to null and is flagged, not imputed; the crisis then has a five-attribute score instead of six.
- **Plan-less active crises.** Reported separately ("possible blind-spot" list), not merged into the main ranking.
- **Sparse donor counts.** HHI flagged as "single-donor-risk" categorical below two donors; not computed numerically.

10 Limitations and Ethical Considerations

Decision support, not automation. The tool produces rankings; humans decide. A score is a hypothesis to investigate, not a verdict.

Visibility bias. Datasets overrepresent crises with functioning information architectures. We surface this via evidence-age flags but cannot fully correct.

[3] https://cerf.un.org/apply-for-a-grant/underfunded-emergencies
[4] https://reliefweb.int/report/angola/breaking-silence-10-most-under-reported-humanitarian-crises-2023

Profile weights are illustrative. The four profiles in Section 5.6 reflect stylised mandate emphases. No specific institution has endorsed any of them. We frame them as plausible sampling points in weight space, not as donor simulations.

Political sensitivity. Declaring a crisis "contested" or "overlooked" is a claim with political implications. Outputs report numeric facts and attribute contributions, not donor-shaming narratives.

False precision. Gap scores reported to two significant figures; disagreement IQR always shown. Scores of 0.84 and 0.82 are treated as equivalent.

Reproducibility. Data fetches timestamped and versioned. External LLM calls logged.

11 Deliverables and Timeline

Engineering risk is concentrated in Phase 1 data harmonisation (HDX HAPI coverage of HNO, FTS plan-year alignment, sector taxonomy reconciliation) rather than scoring logic. The equity and disagreement axes are computationally cheap once sector-level data is clean.

References

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[3] Sahebjamnia, N., Torabi, S.A. and Mansouri, S.A. (2017). *A hybrid decision support system for managing humanitarian relief chains.* Decision Support Systems, 95, 12–26.

[4] Vargas Florez, J., Lauras, M., Okongwu, U. and Dupont, L. (2015). *A decision support system for robust humanitarian facility location.* Engineering Applications of Artificial Intelligence, 46, 326–335.

[5] Griffith, D.A., Boehmke, B., Bradley, R.V., Hazen, B.T. and Johnson, A.W. (2017). *Embedded analytics: improving decision support for humanitarian logistics operations.* Annals of Operations Research, 283, 247–265.

[6] World Food Programme (2019). *WFP and Palantir partnership.* https://www.wfp.org/news/palantir-and-wfp-partner-help-transform-global-humanitarian-delivery

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[9] CARE International (2024). *Breaking the Silence: The 10 most under-reported humanitarian crises.* https://reliefweb.int/report/angola/breaking-silence-10-most-under-reported-humanitarian-crises-2023

Attribute	Source file / columns	Status
a_1 coverage shortfall	`fts_requirements_funding_global.csv` `countryCode`, `year`, `requirements`, `funding`	Ready
a_2 per-PIN gap	above + `hpc_hno_2025.csv` (`In Need`) + `cod_population_admin0.csv` (`Population`)	Ready
a_3 need intensity	`hpc_hno_2025.csv` / `cod_population_admin0.csv`	Ready
a_4 severity	INFORM Severity Index 1–10 (with a 1–5 category), monthly, from the EU JRC DRMKC. 67 monthly snapshots 2020-09 to 2026-03 consolidated into `Data/Third-Party/DRMKC-INFORM/inform_severity_long.csv`. IPC xlsx retained as a richer single-country	Ready — 5,298 country × month rows across 109 countries; every full-stack priority c

Profile	Coverage	Per-PIN	Intensity	Severity	HHI	Equity
CERF-flavoured	0.30	0.15	0.20	0.25	0.05	0.05
ECHO-flavoured	0.25	0.20	0.15	0.15	0.05	0.20
USAID-flavoured	0.20	0.15	0.15	0.15	0.25	0.10
NGO-consortium	0.20	0.15	0.15	0.10	0.10	0.30

Table 3: Illustrative default profiles. CERF emphasises severity and magnitude; ECHO weights equity higher (matching EU humanitarian mandate emphasis); USAID weights donor-concentration (bilateral-system lens); NGO consortium weights cluster-equity dominant.

	Low disagreement (IQR)	**High disagreement (IQR)**
High $G_{\mathbf{cluster}}$	Consensus-starved-sector — clearest operational target	Contested *and* sector-starved — highest advocacy priority
Low $G_{\mathbf{cluster}}$	Consensus-overlooked, uniformly — CERF UFE candidates	Contested but balanced — reveals donor politics

Table 4: Four-cell typology, the primary qualitative output of the system.

Phase	Deliverable	Scope
Phase 1 (MVP)	Country-level ranking with the six-attribute vector, under a single balanced profile; Overlap@10 vs. CERF UFE; explanation strings.	Core task without the novel axes.
Phase 2 (Novel axes)	Four-profile multi-profile driver; disagreement IQR; four-cell typology; Streamlit dashboard with a typology scatter plot.	Core contribution.
Phase 3 (Bonus)	Temporal chronic/acute decomposition; NL query layer; CARE *Breaking the Silence* validation.	Full bonus.
Final deliverable	Working prototype, technical write-up, demo, gap-scoring logic description, failure-mode discussion.	Per challenge brief.